

FIBER-GRAND Paper I: Phase-2E Proof Supplement

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Abstract

This supplement gives complete proofs for the repaired workshop candidate on exact aggregate-ML decoding of an arbitrary nonempty binary membership-testable code under exactly one uniformly located deletion and independent BSC substitutions. It incorporates the independent-review repairs: a finite-exhaustion case split in the stopping theorem, a monotonicity-and-continuity proof of the high-probability exponent, terminology separating a deletion-alignment component from a fully specified latent history, and a constructive witness showing why strict stopping is required for a complete ML tie set. It also gives a uniform-random-code occupancy benchmark for the finite scoring measurements. No generic path-summation, deletion-distance, threshold, membership-oracle, or insertion-sphere principle is claimed as new.

1 Model and exact aggregate likelihood

Let $X = (X_1, \dots, X_n) \in \{0, 1\}^n$. Independent substitutions form $\tilde{X}_i = X_i \oplus N_i$, where $N_i \sim \text{Bernoulli}(p_s)$ independently and $0 \leq p_s < 1/2$. Exactly one coordinate J is deleted uniformly from $\{1, \dots, n\}$. The observation $Y \in \{0, 1\}^m$ has $m = n - 1$. The code $\mathcal{C} \subseteq \{0, 1\}^n$ is arbitrary and nonempty, has a uniform prior, and is accessed only through membership testing. For $x \in \{0, 1\}^n$, $y \in \{0, 1\}^m$, define

$$d_j(x, y) = \sum_{i=1}^{j-1} \mathbf{1}\{x_i \neq y_i\} + \sum_{i=j+1}^n \mathbf{1}\{x_i \neq y_{i-1}\}. \quad (1)$$

Proposition 1 (Likelihood and sliding recurrence). *For every x, y ,*

$$W(y \mid x) = \frac{1}{n} \sum_{j=1}^n p_s^{d_j(x, y)} (1 - p_s)^{m - d_j(x, y)}. \quad (2)$$

Moreover,

$$d_1(x, y) = \sum_{i=2}^n \mathbf{1}\{x_i \neq y_{i-1}\}, \quad (3)$$

$$d_{j+1}(x, y) = d_j(x, y) - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}, \quad 1 \leq j < n. \quad (4)$$

Thus all alignment components are evaluated in $O(n)$ arithmetic operations.

Proof. Condition on $J = j$. The m survivor comparisons require exactly $d_j(x, y)$ substitutions. The substitution variable on the deleted coordinate is unobserved; summing over its two values contributes $(1 - p_s) + p_s = 1$. This gives the j th term of (2); averaging over the uniform J proves the formula. Moving the hypothesized deletion from j to $j + 1$ removes the suffix comparison (x_{j+1}, y_j) and adds the prefix comparison (x_j, y_j) , proving (4). $\square$

2 Strict deletion-alignment-component reversal

Define the largest deletion-alignment component

$$A_y(x) = \max_j \frac{1}{n} p_s^{d_j(x,y)} (1 - p_s)^{m-d_j(x,y)}. \quad (5)$$

This is after marginalizing the substitution on the deleted coordinate. If a “history” is instead defined to include that latent substitution bit, its largest probability is $(1 - p_s)A_y(x)$ for $0 < p_s < 1/2$, a common factor that leaves all candidate orderings unchanged.

Theorem 2 (Strict component/aggregate-ML reversal). *For $n \geq 4$ and $0 < p_s < 1/2$, set*

$$y = 0^{n-3}10, \quad x_A = 0^{n-3}101, \quad x_B = 0^n, \quad \mathcal{C} = \{x_A, x_B\}.$$

Then $A_y(x_A) > A_y(x_B)$, whereas

$$W(y \mid x_B) > W(y \mid x_A) \iff np_s > 1 + 2p_s^2. \quad (6)$$

At equality in (6), the aggregate scores tie while the component inequality remains strict.

Proof. Deleting the final coordinate of x_A gives y without any survivor substitution, so $A_y(x_A) = n^{-1}(1 - p_s)^m$. Every deletion of x_B leaves 0^m , which differs from y in one coordinate; hence $A_y(x_B) = n^{-1}p_s(1 - p_s)^{m-1} < A_y(x_A)$.

For x_A , mismatch distances 0, 1, 2, 3 have multiplicities 1, 1, 1, $n - 3$, respectively; every alignment of x_B has distance one. Therefore

$$\begin{aligned} W(y \mid x_B) &= p_s(1 - p_s)^{n-2}, \\ W(y \mid x_A) &= \frac{1}{n}[(1 - p_s)^{n-1} + p_s(1 - p_s)^{n-2} + p_s^2(1 - p_s)^{n-3} + (n - 3)p_s^3(1 - p_s)^{n-4}]. \end{aligned}$$

Direct factorization gives

$$\frac{n\{W(y \mid x_B) - W(y \mid x_A)\}}{(1 - p_s)^{n-4}} = (1 - 2p_s)(np_s - 1 - 2p_s^2).$$

The first factor is positive on the stated interval, proving the equivalence and equality case. $\square$

Remark 3 (Operating regime). At $p_s = 0.05$, the strict aggregate reversal requires $0.05n > 1.005$, hence integer $n \geq 21$. The family therefore reverses at the tested $n = 24, 32$, but not at $n = 16$. The theorem claims an explicit strict construction, not novelty of the general path-versus-aggregate distinction.

3 Insertion spheres and discovery shells

For $z \in \{0, 1\}^m$, let $\mathcal{I}(z)$ be the set of length- n binary words formed by one insertion.

Lemma 4 (Classical binary one-insertion sphere). *For every z , $|\mathcal{I}(z)| = m + 2 = n + 1$. For inserted bit b , all distinct outputs arise by inserting b in the first gap and immediately after each symbol of z unequal to b .*

Proof. Moving an inserted b through a run of b does not change the output. A distinct output appears initially and every time the insertion crosses a symbol unequal to b , giving $1 + \#\{i : z_i \neq b\}$. Summing over $b = 0, 1$ gives $m + 2$. The two classes differ in Hamming weight and are disjoint. $\square$

Remark 5. Lemma 4 is classical. Likewise, $h_y(x) = \min_j d_j(x, y)$ is, up to orientation, the known minimum one-deletion alignment distance. The only role below is its immediate relation to the inverse enumeration.

Define

$$h_y(x) = \min_{1 \leq j \leq n} d_j(x, y). \quad (7)$$

At shell s , enumerate every $e \in \{0, 1\}^m$ with $\text{wt}(e) = s$, set $z = y \oplus e$, and generate $\mathcal{I}(z)$.

Proposition 6 (Inverse-enumeration first appearance). *Candidate x is generated by the end of shell s if and only if $h_y(x) \leq s$. Its first generation shell is exactly $h_y(x)$.*

Proof. For alignment j , deleting x_j gives $z = y \oplus e$ with $\text{wt}(e) = d_j(x, y)$, and reinserting x_j at coordinate j reconstructs x . Thus x appears at shell d_j for each alignment and no later than $h_y(x)$. Conversely, any generation of x at shell s has an insertion coordinate j whose deletion returns $y \oplus e$ with $\text{wt}(e) = s$, so $d_j(x, y) = s$ and $h_y(x) \leq s$. $\square$

4 Exact unseen-shell certificate

Assume $0 < p_s < 1/2$. After completing shell $s < m$, define

$$U_s = p_s^{s+1} (1 - p_s)^{m-s-1}. \quad (8)$$

Theorem 7 (Unseen-word bound and complete-tie certificate). *Every word unseen after shell $s < m$ has likelihood at most U_s . If a completely scored incumbent codeword has likelihood strictly greater than U_s , the current incumbent set is the complete global ML tie set. Exhaustion at shell m always returns the exact tie set.*

Proof. By Proposition 6, unseen x has $h_y(x) > s$ and therefore $d_j(x, y) \geq s + 1$ for every j . Since $p_s/(1 - p_s) < 1$, each term of (2) is at most U_s ; their average is no larger. A strict incumbent comparison excludes both an unseen strictly better word and an unseen tie. At shell m , all candidates have appeared. $\square$

Proposition 8 (Strictness is necessary for complete ties). *There are interior channel parameters for which a discovered incumbent equals U_s while an unseen codeword ties it. In particular, let $n = 4$, $m = 3$, $y = 001$, $p_s = 1/3$, $s = 0$, and $\mathcal{C} = \{0010, 0000\}$. Then 0010 is discovered at shell zero, 0000 is unseen after shell zero, and*

$$W(y \mid 0010) = W(y \mid 0000) = U_0 = \frac{4}{27}.$$

Thus replacing $>$ by $\geq$ can return an incomplete ML tie set.

Proof. The alignment-distance vectors are $(2, 2, 1, 0)$ for 0010 and $(1, 1, 1, 1)$ for 0000. The first minimum is zero and the second is one. At $p_s = 1/3$, direct substitution into (2) gives $4/27$ for both; (8) also gives $U_0 = (1/3)(2/3)^2 = 4/27$. $\square$

Remark 9 (Boundary cases). At $p_s = 0$, use the standard Bernoulli-mass convention $0^0 = 1$: a word has positive likelihood exactly when some alignment distance is zero, so every positive-likelihood word is generated at shell zero. At $p_s = 1/2$, every candidate has likelihood 2^{-m} and no strict shell decay exists; complete tie certification therefore requires finite exhaustion.

5 Realization-dependent stopping and finite work

Let T be the number of substitutions among the m surviving symbols, put $\rho = p_s/(1 - p_s)$, and define

$$L_n(p_s) = \min\{\ell \geq 1 : \rho^\ell < 1/n\} = \left\lceil \log_{(1-p_s)/p_s} n \right\rceil + 1. \quad (9)$$

The floor formula remains correct when n is an exact power of the logarithm base because the defining inequality is strict.

Theorem 10 (Stopping shell). *For $0 < p_s < 1/2$, the exact decoder stops no later than*

$$S \leq \min\{m, T + L_n(p_s) - 1\}. \quad (10)$$

This bound is independent of the code beyond nonemptiness and membership access.

Proof. Under the true deletion alignment, the transmitted codeword has mismatch count T . It is therefore discovered by shell T and has aggregate likelihood at least

$$\frac{1}{n} p_s^T (1 - p_s)^{m-T}. \quad (11)$$

Let $s_0 = T + L_n - 1$. If $s_0 \geq m$, finite exhaustion gives $S \leq m = \min\{m, s_0\}$. Suppose instead that $s_0 < m$, so U_{s_0} is defined. Comparing (8) with (11),

$$\frac{U_{s_0}}{n^{-1} p_s^T (1 - p_s)^{m-T}} = n \rho^{L_n} < 1.$$

The running incumbent is at least (11), so the strict certificate fires by shell s_0 . $\square$

$$\text{Let } \text{Vol}(m, s) = \sum_{w=0}^{\min\{m, s\}} \binom{m}{w}.$$

Corollary 11 (Deterministic, confidence, and expected work). *For the realized T and the right-hand side S_0 of (10),*

$$Q_{\text{gen}}, Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, S_0), \quad Q_{\text{score}} \leq Q_{\text{mem}}. \quad (12)$$

A history-pair generator attempts $2n \text{Vol}(m, S_0)$ insertion pairs before global deduplication. If τ_δ is the least integer with $\mathbb{P}[T \leq \tau_\delta] \geq 1 - \delta$, then with probability at least $1 - \delta$,

$$Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, \min\{m, \tau_\delta + L_n - 1\}). \quad (13)$$

Moreover,

$$\mathbb{E} Q_{\text{mem}} \leq (n + 1) \sum_{t=0}^m \binom{m}{t} p_s^t (1 - p_s)^{m-t} \text{Vol}(m, \min\{m, t + L_n - 1\}). \quad (14)$$

Proof. Shells through S_0 contain $\text{Vol}(m, S_0)$ error vectors, each with exactly $n + 1$ distinct one-insertion outputs. Global deduplication can only reduce membership calls, and only members are completely scored. Since $T \sim \text{Binomial}(m, p_s)$, monotonicity in T gives (13), and conditioning on T gives (14). $\square$

6 Standard high-probability entropy corollary

Let $h_2(u) = -u \log_2 u - (1 - u) \log_2 (1 - u)$.

Theorem 12 (Oracle/query exponent). *Fix $0 < p_s < 1/2$ and $c > 0$. For all sufficiently large n , with probability at least $1 - n^{-c}$,*

$$Q_{\text{mem}}, Q_{\text{score}} \leq 2^{n(h_2(p_s) + o(1))}. \quad (15)$$

If $|\mathcal{C}_n| = 2^{nR + o(n)}$ with $R > h_2(p_s)$, then on the same event exhaustive codeword-wise ML performs at least $2^{n(R - h_2(p_s) - o(1))}$ times as many complete candidate-likelihood evaluations.

Proof. Put

$$\epsilon_n = \sqrt{\frac{c \ln n}{2m}}, \quad a_n = p_s + \epsilon_n + \frac{L_n - 1}{m}.$$

Hoeffding's inequality gives $\mathbb{P}[T > m(p_s + \epsilon_n)] \leq e^{-2m\epsilon_n^2} = n^{-c}$. On the complementary event, Theorem 10 implies $S/m \leq a_n$. Since $\epsilon_n = o(1)$ and $L_n = O(\log n)$, $a_n = p_s + o(1)$; for all sufficiently large n , $a_n < 1/2$. Hamming-ball volume and h_2 are increasing on $[0, 1/2]$, so

$$\text{Vol}(m, S) \leq 2^{mh_2(a_n)}.$$

By continuity, $h_2(a_n) = h_2(p_s) + o(1)$, while $m/n \rightarrow 1$ and $\log_2(n+1)/n \rightarrow 0$. Combining with (12) yields (15). Dividing $|\mathcal{C}_n|$ by Q_{score} gives the stated exhaustive-scoring ratio. $\square$

Remark 13. The value $h_2(p_s)$ and the typical-volume mechanism are classical for BSC guesswork/GRAND. The content here is only that the exact synchronization decoder inherits this upper exponent after the channel-specific stopping theorem and a polynomial one-deletion factor. This is not an expected GRAND guesswork identity and not a lower bound for all exact decoders.

7 Uniform-random-code occupancy benchmark

Proposition 14 (Exact fixed-set occupancy). *Let the ambient space have size $N = 2^n$. Fix a query set A of size G containing x_0 . Choose $\mathcal{C}$ uniformly among all M -subsets of the ambient space conditioned on $x_0 \in \mathcal{C}$. Then*

$$\mathbb{E}[|A \cap \mathcal{C}| \mid A, x_0] = 1 + (G - 1) \frac{M - 1}{N - 1}. \quad (16)$$

Proof. Write $|A \cap \mathcal{C}| = 1 + \sum_{x \in A \setminus \{x_0\}} \mathbf{1}\{x \in \mathcal{C}\}$. For each such x , symmetry under the conditioned uniform choice gives $\mathbb{P}[x \in \mathcal{C} \mid x_0 \in \mathcal{C}] = (M - 1)/(N - 1)$. Linearity of expectation proves the formula. $\square$

Remark 15 (Interpretation). This is an exact occupancy identity for a fixed query set and random codebook. It is used only as a first-order benchmark. It does not assert that CRC or Hamming codes are random subsets, that the query set is independent of the code, or that a conditional median equals this expectation. For $n = 32$ and the shell-one generation cap $G = 1056$, the predictions are 1.52, 5.12, and 17.48 for $k = 21, 24, 26$, close to the observed median complete-score counts 1.5–3, 4, and 13.5.

8 Claim boundaries after independent review

The generic aggregate likelihood, path-versus-string distinction, minimum deletion-alignment metric, membership-oracle interface, insertion-sphere size, threshold logic, and $h_2(p_s)$ typical volume are prior art or standard consequences. The repaired contribution claims only:

1. the explicit strict binary reversal family and exact threshold;
2. the closed-form aggregate unseen-shell bound U_s for this channel;
3. complete-tie exact certification using that bound;
4. the code-independent realization-dependent stop $S \leq T + L_n - 1$;
5. finite-length exact evidence and a transparent operation-count separation;
6. the occupancy benchmark as an explanatory model, not a universal complexity theorem.

No theorem asserts linear-code structure, unit-cost membership, unique ML decoding, instance optimality, universal latency superiority, or performance under more than one edit.